Minh **Phan**

Leuven, Belgium

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Summary

Biosystem scientist and data-driven process engineer with 6+ years of experience spanning industrial food manufacturing and applied research. Specialized in modelling thermophysical processes and predicting shelf-life kinetics of bio-products, and optimising cold-chain systems to preserve product quality and save energy cost. Demonstrated measurable results including productivity gains up to 30 % in industrial operation settings, a 25 % extension of fruit shelf life under market conditions and a 8 % reduction in energy consumption of food cold chain at scale.

Skills

Transferable skills	Project Management, Leadership, Coaching, Negotiation, Communication, Presentation, Reporting
Process Engineering	Process Design (PFD/P&ID), Process Optimization, System Analysis, Plant Layout Design, Scale-up
Quality Management	HACCP, GMP, Statistical Process Control (SPC), Kaizen, PDCA, 5S
Data Literacy & Analytics	Experiment Design, Data Analysis, Statistical Modeling, Data Visualization
Modeling & Simulation	Mathematical Modeling, Model Calibration & Validation, Sensitivity Analysis, Identifiability Analysis
Programming & Tools	Python (advanced), MATLAB (intermediate), R (beginner), Julia (beginner), JMP, EcosimPro/Modelica

Experience

Applied Research Scientist — KU Leuven, Belgium

Oct. 2022 - Present

- Improved quality control and decision-making in fruit storage systems by developing and validating mathematical models of physiological behaviors and fruit quality from pilot to industrial scale of 'Conference' pear.
- Designed a cost-efficient, user-friendly method for estimating storage energy at the industrial scale, supporting facility-wide adoption.
- Quantified an 8 % potential reduction in storage energy use from innovative technology, providing evidence for cost–benefit evaluation and industrial investment decisions.
- Achieved a 2-day extension in fruit market shelf life using innovative storage technology, compared to commercial practice.
- Designed experiments for testing hypotheses, and analyzed complex datasets with Python
- Worked proactively with interdisciplinary academic and industry teams on modeling, data analysis, and technical problem-solving.
- Supervised masters students, managing their thesis scope, timelines, and technical direction.

Chemical Engineer — AJINOMOTO, Vietnam

Jan. 2018 - Jun. 2019

- Led technology transfer, factory innovation, and CapEx projects, coordinating across Production, QC/QA, and R&D departments.
- Increased 25 % productivity of umami salt line by redesigning purification processes to accommodate a new yeast strain.
- Improved fermentation 10 % productivity of soybean sauce through the oxygen supply optimization.
- Boosted 30 % mayonnaise filling capacity by re-designing a plant layout redesign and setting up a new filling system.
- Developed PFDs/P&IDs, equipment sizing, mass and energy balances, technical specifications, cost estimates, and supplier evaluations for process projects.
- Supported installation, troubleshooting, maintenance, safety, and technical documentation under HACCP, GMP, ISO 22000, and FSSC 22000 standards.
- Applied Kaizen, PDCA, 5S, and root-cause analysis to improve productivity, reduce downtime, and drive continuous improvement across production processes.

Education

Ph.D. in Bioscience Engineering — KU Leuven, Belgium

Oct. 2022 - Present

- *Research project*: Developing a digital twin for the postharvest cold chain of fruits, with a focus on quality and energy efficiency.

M.Sc. in Food Technology — KU Leuven and Ghent University, Belgium

Sep. 2020 - Jul. 2022

Magna Cum Laude

- *Master's thesis*: Non-destructive internal disorder detection of pome fruit using simulated X-ray radiographs and deep learning
  - Developed a deep learning approach (CNN-UNet) for detecting internal disorders in pome fruit from simulated X-ray radiographs.
  - Achieved 94% prediction accuracy.
- *Featured courses*: Thermal and Low-temperature processes of Foods, HACCP and Quality Assurance, Food Rheology, Food Chemistry and Microbiology, Food Fermentation, Postharvest Pest Management and Disease Control, and Project Management

B.Eng. in Food Technology — Ho Chi Minh City University of Technology (VNU), Vietnam

Sep. 2013 - Apr. 2018

First class, Honors Program

- *Bachelor's thesis* focused on evaluating functionality and digestibility of plant-based protein extracted from black beans for food applications.
- *Featured courses*: Food Plant Design, Packaging Technology, Heat Transfer, Mass Transfer, and Fluid Mechanics

Honors & Awards

2022-2026 **PhD Scholarship under the ENOUGH Project (EU Horizon 2020),** KU Leuven *Belgium*

2020-2022 **VLIR-UOS Full Master's Scholarship,** Vlaamse Interuniversitaire Raad *Belgium*

Publication Highlights

An integrated dynamic model of ethylene biosynthesis and respiration of Conference pear (Pyrus communis) during storage and shelf life

***Hoang Minh Phan**, Bert Verlinden, Pieter Verboven, Maarten Hertog, Bart Nicolai*

Postharvest Biology and Technology, 2026

Improving energy efficiency in pear storage through dynamic controlled atmosphere

***Hoang Minh Phan**, Bert Verlinden, Maarten Hertog, Pieter Verboven, Bart Nicolai*

International Journal of Refrigeration, 2026

Non-destructive internal disorder segmentation in pear fruit by X-ray radiography and AI

*Astrid Tempelaere, **Hoang Minh Phan**, Tim Van De Looverbosch, Pieter Verboven, Bart Nicolai*

Computers and Electronics in Agriculture, 2023

Extracurricular Activity

Social Media Manager

Belgium

MeBioS Postharvest Group, KU Leuven *2024 - Present*

- Managed LinkedIn page of *MeBioS Postharvest Group KUL* to successfully attract more than 1000 followers.
- Prepared and published engaging contents, including research highlights, event/achievement announcements, and industry news, to enhance the group's online presence.
- Guided and supported research team members in translating technical and complex research outcomes into accessible, audience-focused content to improve engagement and outreach.

Teaching Assistant

Belgium

MeBioS Postharvest Group, KU Leuven *2024 - 2027*

- Organized and delivered teaching contents (exercise sessions) on *Low-temperature Processing of Foods* for approximately 20 students, focusing on heat transfer and refrigeration principles.
- Guided students in applying theoretical concepts and mathematical models to practical engineering problems.
- Explained complex technical concepts through structured, step-by-step instruction, and interactive Q&A to facilitate learning process.
- Evaluated students' technical reasoning and problem-solving approaches in a course of *Transport Phenomena in Bioscience Engineering*, applying standardized answer keys to ensure accurate, objective, and consistent grading.

Event Organizer

Belgium

MeBioS Postharvest Group, KU Leuven *2022 - 2025*

- Organized annual events from 20 up to more than 100 participants, at both research group and departmental levels, to foster connection, collaboration, and engagement among researchers and research teams.
- Planned event programs, coordinated logistics, and executed activities according to predefined schedules and within budget constraints.
- Managed participant communications, providing timely event information and coordinating arrangements to support smooth event execution.

Languages

Vietnamese (Native) **English (Proficient)** **Dutch (A2-B1)**

References

Prof. Dr. Bart Nicolai

Professor of MeBios Postharvest group, KU Leuven

Director of The Flanders Center of Postharvest Technology

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